Zeguan Wu

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Research Interests

Quantum optimization and numerical linear algebra; quantum interior point methods; continuous-variable and bosonic quantum algorithms for differential equations and scientific computing; quantum resource analysis; AI-assisted mathematical research and formal verification of quantum algorithms.

Research Appointments and Experience

University of Pittsburgh Aug 2025–Present
Postdoctoral Researcher, Department of Computer Science Pittsburgh, PA

- Develop continuous-variable and bosonic quantum algorithms for differential equations, scientific computing, and optimization, collaborating with Juan José Mendoza-Arenas, Peyman Givi, and Junyu Liu.
- Develop AI-assisted mathematical research workflows combining language models, Lean 4, numerical validation, and structured review for quantum algorithms and optimization.

Pacific Northwest National Laboratory Jun 2025–Jul 2025
Quantum Computing Internship Richland, WA

- Led benchmarking and gate-level resource analysis of augmented-Lagrangian Quantum Hamiltonian Descent for constrained nonconvex optimization and power-system instances.

Los Alamos National Laboratory Nov 2023–Apr 2025
Graduate Research Assistant Los Alamos, NM

- Developed and analyzed quantum algorithms for structured linear systems and mathematical optimization, including tensor-format QLSA implementations.

Lehigh University Aug 2020–Aug 2025
Doctoral Research, Industrial and Systems Engineering Bethlehem, PA

- Developed and analyzed quantum interior point methods for linear and quadratic optimization, including feasible and infeasible methods, preconditioning, and dual logarithmic barrier methods.

Education

Lehigh University Aug 2020–Aug 2025
Ph.D., Industrial and Systems Engineering Bethlehem, PA

Advisors: Tamás Terlaky and Xiu Yang. Committee: Arielle Carr, Carleton Coffrin, and Michael Saunders.
Dissertation: "Quantum Computing and Quantum Interior Point Methods."

Columbia University Aug 2018–Dec 2019
M.Sc., Operations Research New York, NY

Nanjing University Sep 2014–Jul 2018
B.Sc., Material Physics Nanjing, China

Peer-Reviewed Journal Articles and Conference Papers

P1. "A preconditioned inexact infeasible quantum interior point method for linear optimization." **Zeguan Wu**, Xiu Yang, and Tamás Terlaky. *Computational Optimization and Applications, 93(3):1225–1257, 2026; doi:10.1007/s10589-025-00750-4.*

P2. "Quantum Interior Point Methods: A Review of Developments and an Optimally Scaling Framework." Mohammadhossein Mohammadisiahroudi, **Zeguan Wu**, Pouya Sampourmahani, Adrian Harkness, and Tamás Terlaky. *Frontiers in Industrial Engineering: Purdue Quantum AI, Communications in Computer and Information Science 2724, 123–139, 2026; doi:10.1007/978-981-95-7829-0_11.*

P3. "A quantum dual logarithmic barrier method for linear optimization." **Zeguan Wu**, Pouya Sampourmahani, Mohammadhossein Mohammadisiahroudi, and Tamás Terlaky. *INFORMS Journal on Optimization, 8(3):185–205, 2026; doi:10.1287/ijoo.2024.0062.*

P4. "An inexact feasible interior point method for linear optimization with high adaptability to quantum computers." Mohammadhossein Mohammadisiahroudi, Ramin Fakhimi, **Zeguan Wu**, and Tamás Terlaky. *SIAM Journal on Optimization, 35(4):2203–2233, 2025; doi:10.1137/23M1589414.*

P5. "Optimal Scaling Quantum Interior Point Method for Linear Optimization." Mohammadhossein Mohammadisiahroudi, **Zeguan Wu**, Pouya Sampourmahani, Jun-Kai You, and Tamás Terlaky. *2025 IEEE International Conference on Quantum Computing and Engineering (QCE), 320–326; doi:10.1109/QCE65121.2025.00044.*

P6. "Improvements to quantum interior point method for linear optimization." Mohammadhossein Mohammadisiahroudi, **Zeguan Wu**, Brandon Augustino, Arielle Carr, and Tamás Terlaky. *ACM Transactions on Quantum Computing, 6(1), Article 8, 1–24, 2025; doi:10.1145/3702244.*

P7. "Network Design for Silent Link User Equilibrium." Xu Chen, **Zeguan Wu**, and Xuan Di. *Research Reports on Computer Science, 2(1):46–55, 2023; doi:10.37256/rrcs.2120232295.*

P8. "An inexact feasible quantum interior point method for linearly constrained quadratic optimization." **Zeguan Wu**, Mohammadhossein Mohammadisiahroudi, Brandon Augustino, Xiu Yang, and Tamás Terlaky. *Entropy, 25(2):330, 2023; doi:10.3390/e25020330.*

P9. "Quantum-enhanced regression analysis using state-of-the-art QLSAs and QIPMs." Mohammadhossein Mohammadisiahroudi, **Zeguan Wu**, Brandon Augustino, Tamás Terlaky, and Arielle Carr. *2022 IEEE/ACM 7th Symposium on Edge Computing (SEC), 375–380, 2022; doi:10.1109/SEC54971.2022.00055.*

Preprints

R1. "Self-Modifying Lean Proof Agents with Verifier-Grounded Benchmark Coevolution." Yuqing Li, **Zeguan Wu**, Yu Gan, and Junyu Liu. *arXiv:2607.17352, 2026.*

R2. "When AI meets quantum information: A comprehensive review." Min Chen, Yu Gan, Xin Jin, Yuqing Li, Junqi Wang, **Zeguan Wu**, Yunfei Wang, Bingzhi Zhang, Priyam Srivastava, Tianlong Chen, Ankit Kulshrestha, Yuan Liu, Juan José Mendoza-Arenas, Kaushik P. Seshadreesan, Sarvagya Upadhyay, Xueyue Zhang, Quntao Zhuang, and Junyu Liu. *arXiv:2607.00365, 2026.*

R3. "Benchmarking and Resource Analysis for Augmented-Lagrangian Quantum Hamiltonian Descent." **Zeguan Wu**, Mingze Li, Muqing Zheng, Meng Wang, Junyu Liu, Samuel Stein, Ang Li, Yousu Chen, and Chenxu Liu. *arXiv:2605.12066, 2026.*

Preprints (continued)

R4. "Infeasibility Aware Large Language Models for Combinatorial Optimization." Yakun Wang, Min Chen, **Zeguan Wu**, Junyu Liu, Sitao Zhang, and Zhenwen Shao. *arXiv:2604.01455, 2026.*

R5. "Artificial Entanglement in the Fine-Tuning of Large Language Models." Min Chen, Zihan Wang, Canyu Chen, **Zeguan Wu**, Manling Li, and Junyu Liu. *arXiv:2601.06788, 2026.*

R6. "Provably Efficient Quantum Algorithms for Solving Nonlinear Differential Equations Using Multiple Bosonic Modes Coupled with Qubits." Yu Gan, Hirad Alipanah, Jinglei Cheng, **Zeguan Wu**, Guangyi Li, Juan José Mendoza-Arenas, Peyman Givi, Mujeeb R. Malik, Brian J. McDermott, and Junyu Liu. *arXiv:2511.09939, 2025.*

R7. "Stab-QRAM: A Clifford-Only Quantum Oracle for Affine Boolean Data." Guangyi Li, Yu Gan, **Zeguan Wu**, Xueyue Zhang, Zheshen Zhang, and Junyu Liu. *arXiv:2509.26494, 2025; version 3, 2026.*

R8. "An Efficient Quantum Algorithm for Linear System Problem in Tensor Format." **Zeguan Wu**, Sidhant Misra, Tamás Terlaky, Xiu Yang, and Marc Vuffray. *arXiv:2403.19829, 2024.*

Accepted Workshop Papers

W1. "Machine-Checked Formalization of the HHL and QSVT Quantum Algorithms in Lean 4." Cross Vargas, Bo Zeng, and **Zeguan Wu**. *Accepted for poster presentation, MATH-AI Workshop, NeurIPS 2026; OpenReview.*

W2. "Claim-Centered Research Records for AI-Assisted Theoretical Research." Yuqing Li, Yu Gan, **Zeguan Wu**, Tianao Wu, and Junyu Liu. *Accepted for poster presentation, MATH-AI Workshop, NeurIPS 2026; OpenReview.*

Conference Abstracts

A1. "An Almost-exact Quantum Interior Point Method for Linear Optimization Achieving Optimal Scaling." Mohammadhossein Mohammadisiahroudi, **Zeguan Wu**, Pouya Sampourmahani, Adrian Harkness, Jun-Kai You, and Tamás Terlaky. *APS Global Physics Summit, March 20, 2026, abstract MAR-Z09.5.* Presenter: Mohammadhossein Mohammadisiahroudi.

Conference Presentations and Posters

1. "AI-Assisted Discovery for Quantum Optimization Algorithms," MOPTA 2026, Bethlehem, PA, August 19, 2026.
2. "Continuous-Variable Quantum Algorithms for Dissipative Dynamics: From PDEs to Optimization." SIAM Annual Meeting, Cleveland, OH, July 7, 2026; INFORMS Optimization Society Conference, Atlanta, GA, March 22, 2026.
3. "Quantum Linear Algebra: from Optimization to Differential Equation," 27th Conference of the International Linear Algebra Society (ILAS 2026), Blacksburg, VA, May 18, 2026.
4. "Quantum interior points methods for linear optimization," SIAM-NNP Annual 2024, Rochester, NY.
5. "A quantum dual logarithmic barrier method for linear optimization." MOPTA 2024, Bethlehem, PA; SIAM Annual 2024, Spokane, WA.
6. "A detailed implementation of QLSA for linear system problem in tensor format," INFORMS Optimization Society Conference 2024, Houston, TX.
7. "An inexact feasible quantum interior point method for linear and quadratic optimization." INFORMS Annual 2023, Phoenix, AZ; MOPTA 2023, Bethlehem, PA.
8. "An inexact feasible quantum interior point method for linearly constrained quadratic optimization." IISE Annual 2023, New Orleans, LA; Mid-Atlantic NA-Day 2022, Philadelphia, PA.
9. "A preconditioned quantum interior point method for linear optimization." INFORMS Annual 2022, Indianapolis, IN (flash talk); ICCOPT & MOPTA 2022, Bethlehem, PA (poster).

Teaching Experience

Teaching Assistant, Lehigh University

- ISE 111: Engineering Probability, Spring 2023.
- ISE 406: Introduction to Mathematical Optimization, Fall 2022.
- ISE 305/404: Simulation, Spring 2022.
- ISE 365/465: Applied Data Mining, Fall 2021.

Summer School Tutorial

- Optimization Laboratory Tutorial, Gene Golub SIAM Summer School, 2023.

Undergraduate Research Mentoring

University of Pittsburgh

- Cross A. Vargas.
- Ben Hoffman.

Academic Service

- Co-organizer: "Quantum Computing: Algorithms and Applications" (Parts I–II), SIAM Annual Meeting, Cleveland, OH, July 7, 2026.
- Session Chair: MOPTA (2024); SIAM Annual Meeting (2024); INFORMS Optimization Society Conference (2024); SIAM-NNP Annual Meeting (2023–2024); INFORMS Annual Meeting (2023–2024); IISE Annual Conference (2023).
- Reviewer: IEEE Transactions on Quantum Engineering; European Journal of Operational Research; Journal of Computational Science; Optimization and Engineering; Scientific Reports; IEEE Quantum Week workshop.
- Vice President, Lehigh University INFORMS Student Chapter, 2022–2023; staff, Gene Golub SIAM Summer School, 2023; volunteer, ICCOPT, 2022.

Awards and Professional Development

- NSF Early Career Funding for ILAS Conference, 2026.
- Los Alamos National Laboratory Quantum Computing Summer School Fellowship, 2023.
- Rossin Professional Development Program, 2023.

Technical Skills

Quantum computing	Qiskit; quantum linear system algorithms; quantum interior point methods; Hamiltonian simulation; block encoding; amplitude amplification; phase estimation; QRAM/QROM; continuous-variable algorithms.
AI-assisted research	Lean 4; proof-agent workflows; language-model evaluation; formal and numerical verification.
Scientific computing	Python; MATLAB; C++; GAMS; CPLEX; Gurobi; AMPL; SQL; Git.

References

- Tamás Terlaky, Industrial and Systems Engineering, Lehigh University.
- Xiu Yang, Industrial and Systems Engineering, Lehigh University.
- Marc Vuffray, Los Alamos National Laboratory.
- Junyu Liu, Computer Science, University of Pittsburgh.